

WEEK – 4, 5

Introduction to Classes, objects, constructors, methods, parameter passing, overloading constructors and methods.

Sample Program:

3. Classes and Objects CO1

Design a “farm animals” java application with the details of animals like cow, pig, horse.

Consider the following details like where they stay, what they eat, the sound they make by using classes and objects.

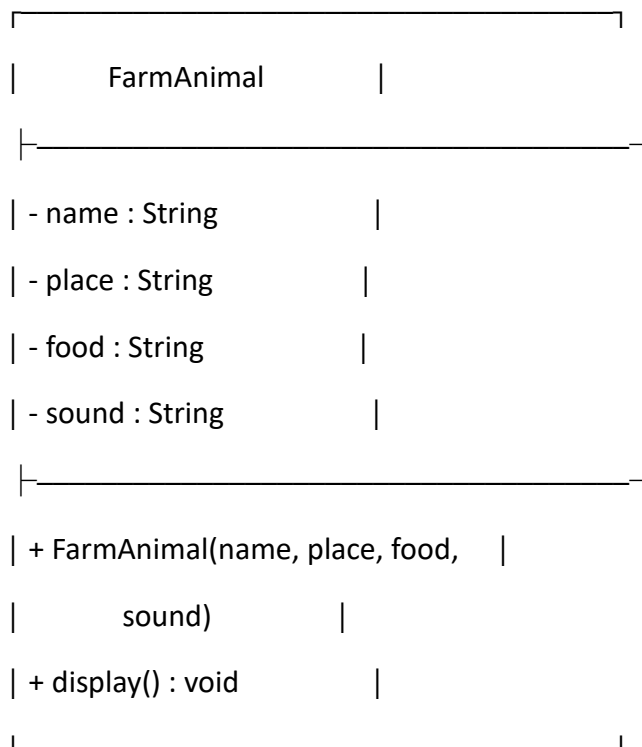
1. Steps of the Program

1. Create a class named FarmAnimal.
2. Declare four variables:
 - name
 - place
 - food
 - sound
3. Create a parameterized constructor FarmAnimal().
4. Use this keyword to initialize the instance variables.
5. Create a display() method to print animal details.
6. Create the FarmAnimals class containing the main() method.
7. Create a cow object and pass its details to the constructor.
8. Create a pig object and pass its details to the constructor.
9. Create a horse object and pass its details to the constructor.
10. Call display() for the cow object.
11. Call display() for the pig object.
12. Call display() for the horse object.
13. Display the details of all three animals.
14. End the program.

2. Overall Algorithm

1. **Start**
2. Define the FarmAnimal class.
3. Declare name, place, food, and sound.
4. Define a parameterized constructor.
5. Initialize the animal details using constructor parameters.
6. Define the display() method.
7. Create a cow object with its name, place, food, and sound.
8. Create a pig object with its name, place, food, and sound.
9. Create a horse object with its name, place, food, and sound.
10. Call display() for the cow.
11. Call display() for the pig.
12. Call display() for the horse.
13. Display all animal details.
14. **Stop.**

3.UML Diagram :



```

    |
    creates objects

    /   |   \
    /   |   \
Cow   Pig   Horse
```

4.Code :

```
class FarmAnimal {

    String name;

    String place;

    String food;

    String sound;


    // Constructor

    FarmAnimal(String name, String place, String food, String sound) {

        this.name = name;

        this.place = place;

        this.food = food;

        this.sound = sound;

    }


    // Method

    void display() {

        System.out.println("Animal : " + name);

        System.out.println("Where it stays : " + place);

        System.out.println("What it eats : " + food);

        System.out.println("Sound : " + sound);

        System.out.println();

    }

}
```

```
}
```

```
public class FarmAnimals {  
    public static void main(String[] args) {  
  
        // Creating objects  
        FarmAnimal cow = new FarmAnimal(  
            "Cow", "Cowshed", "Grass", "Moo");  
  
        FarmAnimal pig = new FarmAnimal(  
            "Pig", "Pigsty", "Grains", "Oink");  
  
        FarmAnimal horse = new FarmAnimal(  
            "Horse", "Stable", "Grass", "Neigh");  
  
        // Calling method  
        cow.display();  
        pig.display();  
        horse.display();  
    }  
}
```

5.Output :

```
PS C:\Users\yabhi\abhi.java> ^C
PS C:\Users\yabhi\abhi.java>
PS C:\Users\yabhi\abhi.java> c::; cd 'c:\Program Files\Java\jdk-9.0.4\bin'
howCodeDetailsInExceptionMessages' '-cp
ava\jdt_ws\abhi.java_92e0559b\bin' 'Far
Animal : Cow
Where it stays : Cowshed
What it eats : Grass
Sound : Moo

Animal : Pig
Where it stays : Pigsty
What it eats : Grains
Sound : Oink

Animal : Horse
Where it stays : Stable
What it eats : Grass
Sound : Neigh
```